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## EDUCATION

### Massachusetts Institute of Technology (MIT)

Class of 2028

- Candidate for B.S., Electrical Engineering with Computing (6-5) & Business Analytics (15-2) | GPA: 4.4/5
- **Relevant Coursework:** Analog Electronics Lab, Semiconductor Electronic Circuits, Signal Processing, Probability and Stats, Optimization Methods in Business Analytics, System Dynamics, Financial Accounting

## EXPERIENCE

### ABS Group - Digital Research Intern, Corporate Technology

June 2026-August 2026

- Built a Python-based company discovery and market segmentation tool: given a target industry (e.g., maritime robotics), the program identifies companies via web scraping and automatically clusters them into market segments using a locally hosted LLM (Ollama)
- Researched photogrammetry as an inspection technology - methodology, software toolchains, strengths, and limitations - producing the technical reference for a presentation to another corporate division; fielded technical questions during delivery
- Surveyed sensor technologies, software platforms, and emerging methods for automated weld inspection to inform team procedural recommendations
- Cataloged and organized robotics-related industry standards (ISO and others) into a structured reference guide for future team use

## RESEARCH EXPERIENCE

### High Voltage Research Laboratory(HVRL) - Temperature Sensing System

September 2025 - May 2026

- Led system architecture and integration, including design of data formats, unique sensor ID scheme, and concentrator logic
- Prototyped a Raspberry Pi 5 software stack: sensor drivers, data collector, time-stamping, storage (CSV + SQLite), and web UI visualization scripts
- Authored a complete system overview and user guide documenting the full software stack for future researchers

## PROJECTS

### EMG Muscle Activity Detection Circuit

Spring 2026

- Designed the signal acquisition and processing chain of a fully analog EMG detection system (team of 2): distributed  $\sim 108$  dB ( $\approx 240,000\times$ ) of gain across a difference-amplifier front end, active filtering, and precision full-wave rectification to convert  $\mu$ V-level biopotentials into a 0-9 V contraction envelope
- Implemented aggressive band-limiting on a single 9 V supply - Sallen-Key high-pass, active Twin-T 60 Hz notch ( $>30$  dB measured rejection), MFB low-pass - to suppress motion artifacts, electrode drift, and mains pickup
- Simulated in LTSpice; built and characterized on breadboard with oscilloscope validation of full signal chain

### Bayesian Model of US Refining Margin (3-2-1 Crack Spread)

Summer 2026

- Built an end-to-end pipeline (EIA, FRED, and futures APIs through posterior diagnostics) modeling refiner margin against inventory and utilization fundamentals over 500 weeks of data, 2015 to 2025
- Identified an asymmetric inventory response: product deficits widen margin several times more than equivalent surpluses compress it, replicated across two independent surplus regimes

### Adjustable Linear Power Supply | Personal Project | *In Progress*

Expected August 2026

- Designing a 0-15V, 2A adjustable bench supply with current limiting and thermal protection; working on design including topology selection, component-level calculations, and LTSpice simulation

## SKILLS

**Circuit Design & Simulation:** Cadence Virtuoso, LTSpice **Programming:** Python, C, Arduino, Linux  
**Lab:** Oscilloscope, function generator, soldering, circuit prototyping and debugging